INCH-POUND

MIL-P-70507 (AR)
AMENDMENT 3
31 October 1990
SUPERSEDING
AMENDMENT 2
3 April 1989

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, XM864
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-P-70507 (AR), dated 28 December 1984, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Add the designation "(M864)" in the title and change XM864 to "M864" wherever it appears in the specification.

PAGE 3

* 3.3: Delete in its entirety and add the following new paragraph:

"3.3 Strength of heat seal. The charge bag shall withstand application of a static load, at the seam, of 2.5 pounds per linear inch, minimum without peel-back of the heat seal and without tensile failure of the area adjacent to the heat seal."

PAGE 4

* 3.5: Delete 3.5.g "separated igniter components" and substitute in 3.5.g "Examine the base burner assembly for:

1. Missing components
2. Misassembled components
3. Igniter cup loading assemblies facing the wrong way (inverted orientation)
4. Foreign matter present

* Add "3.5.h Presence of gap between base and body"

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PAGE 5

* 4.3.1: Delete the table in its entirety and substitute the following:

"

Part Description	Drawing	Quantity
Projectile	9381131	12
Steel Shim	9215397	25
Expulsion Charge Assembly	12934964	50
Obturator	1054290	5
Cup, Expulsion	9391057	25
Charge Loading Assembly	12934963	25
Bag, Charge Assembly	12934962	25
Cloth, Laminated Basic Cut	12934961	25"

PAGE 6

* Table I: Delete in its entirety and substitute the following:

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Table I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)
AMENDMENT 3

PARAGRAPH	TITLE: Projectile and Components		SHEET 1 OF 2		DRAWING NUMBER
					NEXT HIGHER ASSEMBLY: See below
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile (Dwg. 9381131)				
	Examination for defects	12	-	3.1	4.4.2.14
	Ballistic functioning	12	-	3.4	4.4.3.1
	Expulsion Charge Assembly (Dwg. 12934964)				
	Examination for defects	50		3.2	4.4.2.4
	Steel Shim (Dwg. 9215397)				
	Examination for defects	25		3.2	4.4.2.7
	Bag Charge Assembly (Dwg. 12934962)				
	Examination for defects	25		3.2	4.4.2.1
	Cover, Expulsion Charge (Dwg. 9272022)				
	Examination for defects	25		3.2	4.4.2.2
	Charge Loading Assembly (Dwg. 12934963)				
	Examination for defects	25		3.2	4.4.2.3

NOTES:

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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* Table I.: Delete the table in its entirety and substitute the following:

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Table I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)
AMENDMENT 3

PARAGRAPH	TITLE: Projectile and Components		SHEET 2 OF 2		DRAWING NUMBER
					NEXT HIGHER ASSEMBLY: See below
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Obturator (Dwg. 10542907)				
	Examination for defects	(a)		3.2	4.4.2.16
	Forward Plate (Dwg. 9381111)				
	Examination for defects	10		3.2	4.4.2.24

NOTES: (a) Five (5) samples from each mold cavity.

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PAGE 9

* 4.4.1: Delete item d and substitute the following:

"d. Base burner assemblies from one finally accepted lot interfix number from one manufacturer."

Delete the following restrictions:

"g. from one lot interfix number

h. from one lot interfix number"

Delete item k and substitute the following:

"k. Expulsion charge assemblies from one lot interfix number from one manufacturer."

Add the following:

"o. Expulsion cups from one lot interfix number from one manufacturer."

PAGE 10

* 4.4.2.1: Delete in its entirety and add the following paragraph.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)
AMENDMENT 3

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Bag, Charge		9272021
			NEXT HIGHER ASSEMBLY 9272020

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
~~Critical~~	~~None defined~~				
Major					
101	Length of bag		0.40%	3.2	Gage
102	Distance to end heat seal, max.		0.40%	3.2	Gage
103	Distance to side heat seal		0.40%	3.2	Gage
104	Minimum width of both heat seals over entire length		0.40%	3.2	Gage
105	Heat seal incomplete (2 places)		0.40%	3.2	Visual
106	Strength of heat seal	(a)		3.2	4.5.3
107	Location of eyelet		0.40%	3.2	Gage
108	Location of end stitching		0.40%	3.2	Gage
109	Distance to side stitching		0.40%	3.2	Gage
Minor					
201	Stitching incomplete (2 places)		0.65%	3.2	Visual
202	Fold width eyelet end		0.65%	3.2	Gage
203	Fold misalignment max.		0.65%	3.2	Gage
204	Thickness of assembled eyelet		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES: (a) Sample 20 per lot, reject on one (1) or more defectives.

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* 4.4.2.2: Delete in its entirety and add the following paragraph:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)
AMENDMENT 3

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Cover, Expulsion Charge		9272022
			NEXT HIGHER ASSEMBLY 129349263

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of rim		0.40%	3.2	Gage
102	Thickness of rim, minimum		0.40%	3.2	Gage
103	Total height		0.40%	3.2	Gage
104	Angle between rim and angled portion		0.40%	3.2	Gage
105	Thickness of angled portion, minimum		0.40%	3.2	Gage
106	Height of rim		0.40%	3.2	Gage
107	Location of tabs		0.40%	3.2	Gage
108	Distance between tab and angled portion (2 places)		0.40%	3.2	Gage
109	Radius missing		0.40%	3.2	Visual
110	Crack or puncture		0.40%	3.2	Visual
111	Burr		0.40%	3.2	Visual
112	Hole diameter under minimum		0.40%	3.2	Gage
Minor					
201	Hole diameter over maximum		0.65%	3.2	Gage
202	Hole true position		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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* 4.4.2.3: Delete in its entirety and add the following paragraph:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)
AMENDMENT 3

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.3	Charge Loading Assembly	1	1	9272020
				NEXT HIGHER ASSEMBLY 9272019

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pull wire or bag loose with respect to cover		0.40%	3.2	Visual/Manual
102	Cover damaged		0.40%	3.2	Visual
103	Clinch of stud improper		0.40%	3.2	Visual
104	Bag diameter		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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* 4.4.2.4: Delete in its entirety and add the following paragraph:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)
AMENDMENT 3

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.4	Expulsion Charge Assembly	1	1	12934964
				NEXT HIGHER ASSEMBLY: 9381131

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Leak test	(a)		3.2	4.5.2
102	Weight of propellant charge incorrect		0.40%	3.2	Gage
103	Bag seal incomplete (3 places)		0.40%	3.2	Visual
104	Moisture and total volatiles content of propellant	(a)		3.2	4.5.1
105	Bag damaged.		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

(a) Sample in accordance with applicable test method paragraph.

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PAGE 20

* 4.4.2.11: Major 104, change "sealant" to "adhesive", change AQL "0.40%" to 100%".

PAGE 21

* 4.4.2.12: Special Defect a, change "Gage" to "Gage/4.5.14".

* Add the following minor defect:

"Minor 202	Sealing compound not present between projectile body and base (360)	.65%	3.2	Visual"

Delete Major Defect 101 and substitute the following:

"Major 101	Disassembly torque test of base burner assembly	100%	3.2	4.5.6"

Delete Major Defect 103 in its entirety.

PAGE 27

* 4.4.2.17: Delete in its entirety.

PAGE 28

* 4.4.2.18: Delete in its entirety.

PAGE 29

* 4.4.2.19: Delete in its entirety.

PAGE 30

* 4.4.2.20: Delete in its entirety.

PAGE 31

* 4.4.2.21: Delete in its entirety and substitute the following:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)
AMENDMENT 3

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630555
4.4.2.21	Washer, Expulsion				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
103	Thickness		0.40%	3.2	Gage
104	Angle of tape (two places 180 apart)		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.2	Visual

NOTES:

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4.4.2.22: Replace existing paragraph with new 4.4.2.22 as follows:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)
AMENDMENT 3

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.22	Cup, Expulsion		9391057
			NEXT HIGHER ASSEMBLY: 9381131

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of thread		0.40%	3.2	Gage
102	Major diameter of thread		0.40%	3.2	Gage
103	True position of largest inside diameter with pitch diameter of threads		0.40%	3.2	Gage
104	True position of smallest inside diameter with pitch diameter of thread		0.40%	3.2	Gage
105	Minimum inside diameter		0.40%	3.2	Gage
106	True position of largest outside diameter with pitch diameter of thread		0.40%	3.2	Gage
107	Length of threaded area		0.40%	3.2	Gage
108	Hardness test (only applies for alternate material)	10	note a	3.2	4.5.13
109	Thickness of bottom		0.40%	3.2	Gage
110	Thickness of sidewall		0.40%	3.2	Gage
111	Overall length		0.40%	3.2	Gage
112	Depth to start of angle		0.40%	3.2	Gage
113	Presence of two slots		0.40%	3.2	Visual
114	Largest outside diameter		0.40%	3.2	Gage
115	Perpendicularity of largest outside diameter		0.40%	3.2	Gage
116	Presence of angle between large & small inside diameters		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES: a. Each heat treat batch (see 6.7). If any cup fails to comply with the applicable requirement, the heat treat batch represented by the sample shall be rejected.

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PAGE 34

4.4.2.24: Major 104:

Change "Concentricity" to "True Position".

PAGE 35

* 4.4.3.1: Delete and substitute the following:

"4.4.3.1 Ballistic Functioning (see 3.4). For first article and every lot produced, the projectile shall be tested per 4.5.10 as follows:

Test Phase	Initial Sample	Retest Sample
Service Pressure	5	5
Safety	2	2

If a projectile failure as defined by 4.4.3.1.3 occurs during testing of the initial sample, another seven round quantity will be fired as described in the retest sample test quantity above. Failure of a lot quantity will occur if two or more projectiles fail to meet the requirements of 4.4.3.1.3."

PAGE 36

* 4.4.3.1.3e. Delete and substitute the following:

"e. Any tow or more projectiles fail to function satisfactorily (major defect)."

PAGE 38

4.5.2: Delete paragraph 4.5.2 but retain 4.5.2.1.

PAGE 39

4.5.2.1: Change dwg. "936255" to "9391003".

* 4.5.3: Delete paragraph in its entirety.

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PAGE 40

* 4.5.9: Delete in its entirety and substitute the following:

"4.5.9 Radiographic inspection of projectile. The projectile assemblies shall be radiographically examined prior to forwarding to the proving ground. Prior to assembly of the base burner assembly, the Government inspector shall indicate the location of the spline by marking the outside of the projectile. In all exposures the radiographic beam shall be focused along the center axis of the projectile and parallel to the face of the forward plate, with the spline ninety degrees (90) to the right in the direction from the source to the projectile. Radiographic exposure shall as a minimum, include all of the base burner assembly, to and including the entire forward plate. If necessary more than one radiograph shall be taken in this view as required to avoid detrimental parallax effects and to observe the conditions specified in 3.5. In the event that the radiograph(s) taken in this view prevent a suitable examination, the projectile shall again be radiographed at a location rotated 90 from the original view, with the spline nearest to the source. Projectiles and radiographs shall be suitably identified to permit cross reference. Radiographs shall be maintained by the contractor until final acceptance of the projectile. If any defect, per paragraph 3.5 is found, it shall be reported to AMSMC-QAR-Q(D). The lot represented by this sample shall be rejected.

4.5.9.1 Radiographic qualifications. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. The personnel performing this examination shall be qualified and certified, Level 2, per MIL-STD-410. In addition, the radiographic examination shall be performed with the technique capable of producing 2 percent sensitivity (radiographic quality level 2-2T) and the photographic density of the film used shall be of a quality which will produce a sharp definition of internal structure."

PAGE 42

* 4.5.12: Delete in its entirety.

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4.5.13: Add new paragraph as follows:

"4.5.13 Hardness test of cup. The cups selected shall be subjected to the hardness test in accordance with the method and procedures of ASTM E 18-67. The load and penetrator shall be in accordance with the hardness scale specified on the applicable drawing. Three readings shall be taken on each cup on the outside surfaces of the closed and approximately 1/4" from the edge and approximately 120 apart. A suitable fixture shall be utilized to provide support under the surface being tested. Such fixtures shall be approved in accordance with 4.4.4. If the average value of the three readings for any cup is less than the value specified in the applicable drawing, that cup shall have failed the requirement. Equipment used for performing this test shall be approved in accordance with paragraph 4.4.4."

* 4.5.14: Add the following new paragraph:

4.5.14 Inspection of gap between projectile body and base. Gap shall be inspected using an approved feeler gage as specified on drawing 9381131. The feeler gage shall be cleaned with a clean dry rag immediately after each inspection such that the gage is free of any base/body gap sealant. No solvent or other substance other than a clean dry rag shall be used to clean the gage. The gap between the projectile body and base shall be inspected within fifteen (15) minutes after the sealant is applied to the shoulder of the base."

PAGE 43

Add new paragraph 6.7 as follows:

"6.7 Heat-treatment batch. A heat-treatment batch is herein defined as cups, of the same type, that have been produced from one lot of material, and heat treated at the same time in the same furnace, quench medium and aging process for each of all phases of the entire heat treatment process. When a continuous-type heat treatment process is used, the quantity of cups processed through the furnace for each four (4) hours production, or portion thereof, shall be considered as a heat treatment batch."

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